

MEREWETHER HIGH SCHOOL
YEAR 12 TRIAL HSC EXAMINATIONS 2000

MATHEMATICS

4 Unit ADDITIONAL PAPER

Time allowed: Three hours plus 5 minutes reading time.

INSTRUCTIONS: *All questions may be attempted
*All questions are of equal value
*In every question all necessary working should be shown -
full marks may not be awarded for answers only.
*Approved silent calculators may be used.
*Standard integrals are printed at the back of this exam. paper.
*Start each question on a new sheet of paper.

Question 1

Marks

a (i) Find $\int \frac{dx}{x \ln x}$

1

(ii) Find $\int \frac{dx}{4 + 3 \cos x}$

2

(iii) Evaluate $\int_{-1}^0 \frac{(x-1)dx}{x^2 + 2x + 2}$

3

b Use integration by parts to evaluate $\int_0^{\frac{1}{2}} \cos^{-1} x dx$

3

c (i) Find A, B and C so that

$$\frac{10}{(3+x)(1+x^2)} = \frac{A}{3+x} + \frac{Bx+C}{1+x^2} \text{ for all } x, x \neq -3$$

3

(ii) Hence evaluate $\int_0^1 \frac{10}{(3+x)(1+x^2)} dx$

3

Question 2 (START A NEW SHEET OF PAPER)

a Find the cube roots of $27i$

4

NOTE: parts (b), (c) and (d) are NOT related

b On an Argand diagram, shade in the region defined by

$$\operatorname{Im}(z) \leq 1 \text{ and } \frac{\pi}{3} \leq \arg(z+i) \leq \frac{\pi}{2}$$

3

c The complex number z and its conjugate $\bar{z}$ satisfy the equation

$$z\bar{z} - 2iz = -3 - 2i$$

Find the possible values of z .

4

d (i) Sketch the graph specified by $|z - 2 - i\sqrt{3}| = \sqrt{7}$

3

(ii) Hence find the maximum value of $|z|$

1

Question 3 (START A NEW SHEET OF PAPER)

a $(1-i)$ is a root of the equation $x^4 - 3x^3 + 3x^2 - 2 = 0$. Find all the other roots.

3

b Consider the cubic equation $P(x) = x^3 + ax + b$

Show that if $a > 0$, then $P(x) = 0$ has exactly one real root.

3

c Sketch the following curves on SEPARATE sets of axes, showing clearly all the main features:

(i) $y = (x+1)(3-x)$

1

(ii) $y = \frac{1}{(x+1)(3-x)}$

2

(iii) $y = \left| \frac{1}{(x+1)(3-x)} \right|$

2

(iv) $y = \log_e(x+1)(3-x)$

4

Question 4 (START A NEW SHEET OF PAPER)

- a (i) Show that the normal to the hyperbola $xy = c^2$ at the point $P\left(ct, \frac{c}{t}\right)$

has the equation $y = t^2x + \frac{c}{t} - ct^3$

2

- (ii) If the normal at P meets the line $y = x$ at N, and the tangent at P meets $y = x$ at T, find the co-ordinates of N and T.

2

- (iii) If O is the origin, prove that $OT \cdot ON = 4c^2$

3

- b The ellipse E has cartesian equation $\frac{x^2}{25} + \frac{y^2}{16} = 1$

- (i) State E's eccentricity, the co-ordinates of its foci S and S' and the equations of its directrices.

3

- (ii) Sketch the curve neatly, showing essential features.

3

- (iii) If P is any point on E, and the length of the interval PS is 2 units, find the length of the interval PS'

2

Question 5 (START A NEW SHEET OF PAPER)

- a (i) Sketch on the number plane the circle $(x-1)^2 + y^2 = 4$, labelling all intercepts on the x and y axes.

2

- (ii) On this diagram shade the region

$$\{(x, y) : (x-1)^2 + y^2 \leq 4\} \cap \{(x, y) : x \geq 0\}$$

1

- (iii) Your shaded region in part (ii) forms the base of a solid with every cross-section perpendicular to the x-axis forming a square, one side of which lies on the base. Find the volume of the solid.

5

- b (i) Given $I_n = \int_0^1 x^n e^{2x} dx$ where n is a positive integer, use integration by parts to show that:

$$I_n = \frac{1}{2} (e^2 - n \times I_{n-1})$$

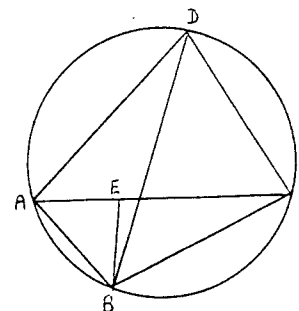
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- (ii) Hence evaluate $\int_0^1 x^3 e^{2x} dx$

3

Question 6 (START A NEW SHEET OF PAPER)

a



The figure above shows a cyclic quadrilateral ABCD with diagonals AC and BD. E is a point on AC such that $\angle ABE = \angle DBC$.

- (i) Prove that:

$$(\alpha) \triangle ABE \sim \triangle DBC$$

2

$$(\beta) \triangle ABD \sim \triangle ECB$$

2

- (ii) Hence prove Ptolemy's Theorem, which is that:

$$BA \times DC + AD \times BC = AC \times BD$$

3

- b If the circular disc with centre (3,0) and radius 2 is rotated about the y-axis, then a doughnut-shaped solid is formed.

- (i) Use the method of cylindrical shells to show clearly that the volume of this solid is given by:

$$V = 4\pi \int_1^5 x \sqrt{4 - (x-3)^2} dx$$

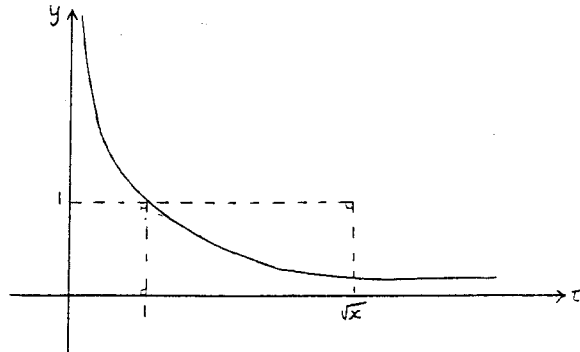
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- (ii) Hence find the volume of the solid.

4

Question 7 (START A NEW SHEET OF PAPER)

a



This diagram shows that $0 < \int_1^{\sqrt{x}} \frac{dt}{t} < \sqrt{x}$, for all $x > 1$

Evaluate this integral, and then use this inequality to show that:

$$\lim_{x \rightarrow \infty} \left(\frac{\ln x}{x} \right) = 0$$

2

b (i) Find in exact form all turning points and points of inflexion

on the curve $y = \frac{\ln x}{x}$,

$$\text{given } \frac{dy}{dx} = \frac{1 - \ln x}{x^2} \text{ and } \frac{d^2y}{dx^2} = \frac{2 \ln x - 3}{x^3}$$

2

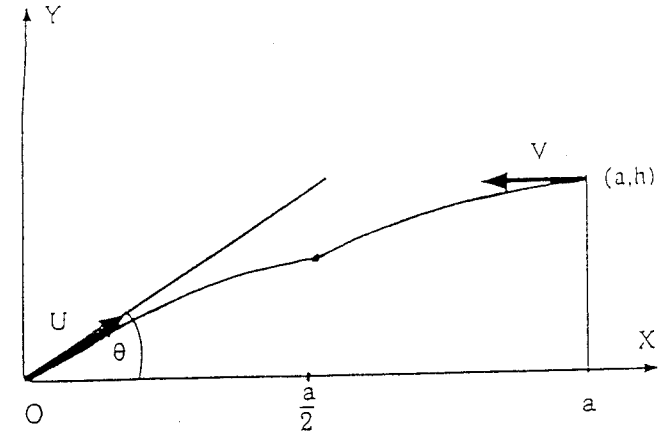
(ii) Sketch $y = \frac{\ln x}{x}$

3

Question 7 continued on the next page.

Question 7 continued

c



A gun is so aimed that the shell it fires strikes a target released simultaneously from an aeroplane flying horizontally towards the gun at a speed of $V \text{ ms}^{-1}$ and at a height ' h ' metres. The aeroplane was at a horizontal distance ' a ' metres from the gun when the target was released, and the shell strikes the target at half this horizontal distance ' a ', as shown on the diagram. The initial velocity of the shell is $U \text{ ms}^{-1}$ and the angle of projection is θ

(i) Show that the equations of motion of the target are:

$$\begin{aligned} \ddot{x} &= -V \\ \dot{x} &= -gt \\ x &= a - Vt \end{aligned} \quad \begin{aligned} \dot{y} &= -gt \\ y &= h - \frac{1}{2}gt^2 \end{aligned}$$

3

(iii) Show that the gun was aimed at a point h metres vertically above the aeroplane at the instant of release, and that

$$U = \frac{V}{\sin \theta} \sqrt{a^2 + 4h^2}$$

5

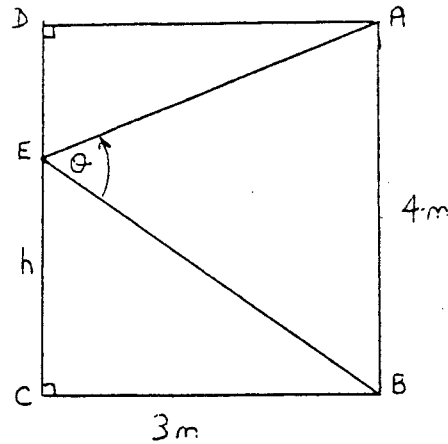
Question 8 (START A NEW SHEET OF PAPER)

- a** (i) $\sin(A+B) = \sin A \cos B + \sin B \cos A$ and $\cos(A+B) = \cos A \cos B - \sin A \sin B$

Prove that $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

2

(ii)



ABCD is a rectangle. AB is 4 metres long and BC is 3 metres. E is a variable point on side CD. Let $\angle AEB$ be θ and EC be h metres in height.

(i) Show that $\tan \theta = \frac{12}{9 - 4h + h^2}$

3

(ii) What value of h makes θ a maximum?

4

- b** On a certain day the depth of water in a bay at high tide was 11m. At low tide, $6\frac{1}{4}$ hours later, the depth of water was 7m. If the next high tide is due at 3.20pm, what is the earliest time that a ship, which needs a depth of at least 10m, can enter the bay? (Assume that the rise and fall of the tide is Simple Harmonic)

6

1 a) (i) $I = \int \frac{dx}{x \ln x}$ let $u = \ln x$
 $du = \frac{1}{x} dx$
 $= \int \frac{du}{u}$
 $= \ln u + c$
 $= \ln(\ln x) + c$

(ii) $I = \int \frac{dx}{4+3\cos x}$ let $t = \tan \frac{x}{2}$
 $= \int \frac{2dt}{1+t^2}$
 $= \int \frac{2dt}{4+3\frac{(1-t^2)}{1+t^2}}$
 $= \int \frac{2dt}{4+4t^2+3-t^2}$
 $= \int \frac{2dt}{t^2+7}$
 $= \frac{2}{\sqrt{7}} \tan^{-1} \frac{t}{\sqrt{7}} + c$
 $= \frac{2}{\sqrt{7}} \tan^{-1} \left(\frac{\tan \frac{x}{2}}{\sqrt{7}} \right) + c$

(iii) $I = \frac{1}{2} \int \frac{(2x-2)dx}{x^2+2x+2}$
 $= \frac{1}{2} \int \frac{(2x+2)dx}{x^2+2x+2} - \frac{1}{2} \int \frac{4dx}{x^2+2x+2}$
 $= \frac{1}{2} \left[\ln(x^2+2x+2) \right] - \int \frac{2dx}{(x+1)^2+1}$
 $= \frac{1}{2} (\ln 2 - \ln 1) - 2 \left[\tan^{-1}(x+1) \right]_{-1}^0$
 $= \frac{1}{2} \ln 2 - 2(\tan^{-1} 1 - \tan^{-1} 0)$
 $= \frac{1}{2} \ln 2 - 2\left(\frac{\pi}{4} - 0\right)$
 $= \frac{1}{2} \ln 2 - \frac{\pi}{2}$

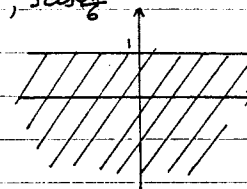
b) $I = \int_0^{\frac{\pi}{2}} \cos^2 x \cdot 1 \cdot dx$
 $\int uv' dx = uv - \int vu' dx$
 $= [\cos^2 x \cdot x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} x \cdot \frac{-1}{\sqrt{1-x^2}} dx$
 $= \left(\frac{1}{2} \cos^{-1} \frac{1}{2} - 0 \right) - \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{-2x dx}{\sqrt{1-x^2}}$
 $= \frac{1}{2} \times \frac{\pi}{3} - \frac{1}{2} \left[\frac{(1-x^2)^{\frac{1}{2}}}{\frac{1}{2}} \right]_0^{\frac{\pi}{2}}$
 $= \frac{\pi}{6} - \left(\sqrt{1-\frac{1}{4}} - \sqrt{1-0} \right)$
 $= \frac{\pi}{6} - \frac{\sqrt{3}}{2} + 1$

c) (i) $\frac{10}{(3+x)(1+x^2)} = \frac{A}{3+x} + \frac{Bx+C}{1+x^2}$
 $\therefore 10 \equiv A(1+x^2) + (3+x)(Bx+C)$
 let $x = -3$ $10 = 10A \Rightarrow A = 1$
 let $x = 0$ $10 = A + 3C \Rightarrow C = 3$
 let $x = 1$ $10 = 2A + 4B + 4C \Rightarrow B = -1$
 $\therefore A = 1, B = -1, C = 3$

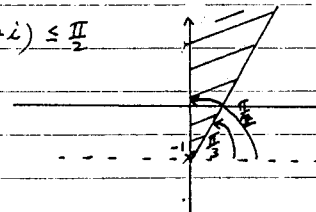
(ii) $I = \int_0^1 \frac{10 \cdot dx}{(3+x)(1+x^2)}$
 $= \int_0^1 \left(\frac{1}{3+x} + \frac{3-x}{1+x^2} \right) dx$
 $= \int_0^1 \left(\frac{1}{3+x} + \frac{3}{1+x^2} - \frac{x}{1+x^2} \right) dx$
 $= \left[\ln(3+x) + 3 \tan^{-1} x - \frac{1}{2} \ln(1+x^2) \right]_0^1$
 $= \ln 4 + 3 \tan^{-1} 1 - \frac{1}{2} \ln 2 - \ln 3 - 3 \tan^{-1} 0$
 $+ \frac{1}{2} \ln 1$
 $= \ln 4 - \ln 2 - \ln 3 + \frac{3\pi}{4}$
 $= \ln \frac{4}{3\sqrt{2}} + \frac{3\pi}{4}$

2. a) Let $x^3 = 27i$
 $x^3 - 27i = 0$
 $x^3 + 27i^3 = 0$
 $(x+3i)(x^2-3ix+9i^2) = 0$
 $(x+3i)(x^2-3ix-9) = 0$
 $x = -3i$ $x = \frac{3i \pm \sqrt{9i^2+36}}{2}$
 $= \frac{3i \pm \sqrt{27}}{2}$
 $= \frac{3i \pm 3\sqrt{3}}{2}$
 $\therefore$ cube roots of $27i$ are $-3i, \frac{3i \pm 3\sqrt{3}}{2}$
 OR $3i \cos \frac{\pi}{6}, 3i \cos \frac{5\pi}{6}, 3i \cos \frac{9\pi}{6}$

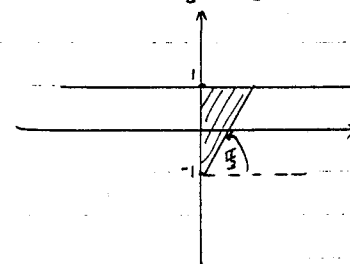
b) $\text{Im}(z) \leq 1$



$\frac{\pi}{3} \leq \arg(z+i) \leq \frac{\pi}{2}$



$\text{Im}(z) \leq 1$ and $\frac{\pi}{3} \leq \arg(z+i) \leq \frac{\pi}{2}$

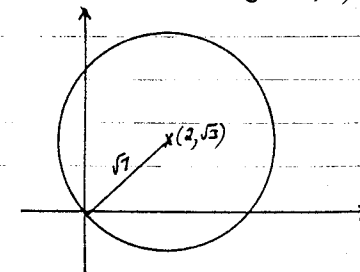


c) $z\bar{z} - 2iz = -3-2i$
 let $z = x+iy$, so $\bar{z} = x-iy$
 $\therefore (x+iy)(x-iy) - 2i(x+iy) = -3-2i$
 $x^2+y^2 - 2ix - 2i^2y = -3-2i$
 $x^2+y^2+2y - 2ix = -3-2i$
 equating reals & imaginaries,
 $x^2+y^2+2y = -3$, $x = 1$
 $y^2+2y+1+3 = 0$

$y^2+2y+4 = 0$
 $y = \frac{-2 \pm \sqrt{4-16}}{2}$
 $= \frac{-2 \pm \sqrt{-12}}{2}$
 $= \frac{-2 \pm 2\sqrt{3}i}{2}$
 $= -1 \pm \sqrt{3}i$
 $\therefore z = 1 + i(-1 \pm \sqrt{3}i)$
 $= 1 - i \pm \sqrt{3}i^2$
 $= 1 \pm \sqrt{3} - i$
 $= 1 - \sqrt{3} - i, 1 + \sqrt{3} - i$

d) $|z - 2 - i\sqrt{3}| = \sqrt{7}$
 circle, centre $2+i\sqrt{3}$, radius $\sqrt{7}$
 Note however $|2+i\sqrt{3}| = \sqrt{2^2+(\sqrt{3})^2} = \sqrt{7}$

$\therefore$ circle passes through $(0,0)$



max value of $|z|$ = max dist of any point on circle from $(0,0)$

$\therefore \max |z| = 2\sqrt{7}$

(3)

3. a) Since $(1-i)$ is a root, so is $(1+i)$
 $\therefore (x-(1-i))(x-(1+i))$ is a factor of $P(x)$

$$\text{ie } (x-1)^2 - i^2 = x^2 - 2x + 2$$

$$\begin{array}{r} x^2 - x - 1 \\ x^2 - 2x + 2 \overline{) x^4 - 3x^3 + 3x^2 + 0x - 2} \\ \underline{x^4 - 2x^3 + 2x^2 + 0x} \\ -x^3 + x^2 + 0x \\ \underline{-x^3 + 2x^2 - 2x} \\ -x^2 + 2x - 2 \\ \underline{-x^2 + 2x - 2} \\ 0 \end{array}$$

$$\therefore \text{eqn becomes } (x^2 - 2x + 2)(x^2 - x - 1) = 0$$

$$x = 1-i, 1+i \quad x = \frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}$$

$$\therefore x = 1 \pm i, \frac{1 \pm \sqrt{5}}{2}$$

$$b) P(x) = x^3 + ax + b$$

$$P'(x) = 3x^2 + a$$

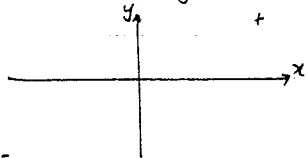
$$\text{If } a > 0, 3x^2 + a > 0 \therefore P'(x) > 0$$

$\therefore P(x)$ is monotonically increasing throughout

$$\text{If } x \rightarrow -\infty, P(x) \rightarrow -\infty$$

$$\text{and If } x \rightarrow \infty, P(x) \rightarrow \infty$$

and $P(x)$ is a polynomial

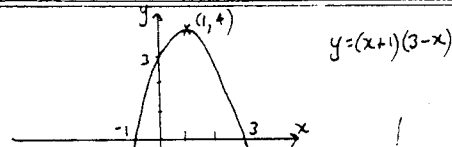


$\therefore P(x)$ must cut x -axis once only

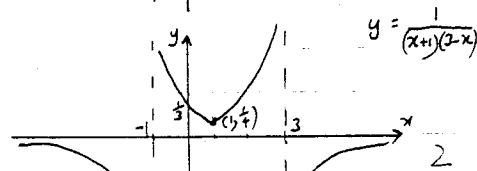
$\therefore P(x) = 0$ has exactly one root if $a > 0$

3

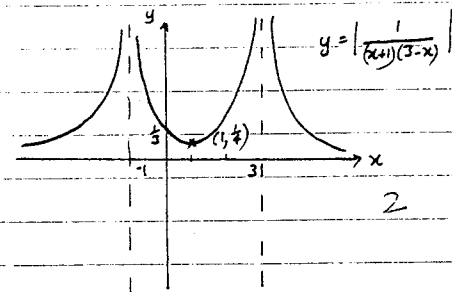
(c) (i)



(ii)

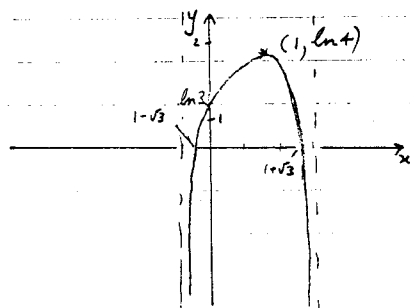


(iii)



(iv)

$$(x+1)(3-x) > 0 \therefore -1 < x < 3$$



4

(4)

$$4 a) \quad xy = c^2$$

$\therefore xy' + y = 0$ by implicit diff.

$$\therefore y' = -\frac{y}{x}$$

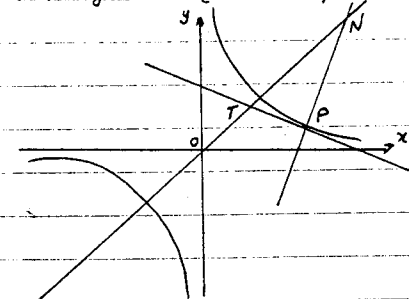
$$\text{at } (ct, \frac{c}{t}) \text{ m of tan} = -\frac{c/t}{ct} = -\frac{1}{t^2}$$

$$\therefore \text{m of norm} = t^2$$

$$y - y_1 = m(x - x_1)$$

$$y - \frac{c}{t} = t^2(x - ct)$$

$$\therefore y = t^2x + \frac{c}{t} - ct^3 \text{ is eqn of norm at P}$$



to find N, let $y = x$

$$\therefore x = t^2x + \frac{c}{t} - ct^3$$

$$(t^2 - 1)x = c(t^2 - \frac{1}{t})$$

$$x = \frac{c(t^4 - 1)}{t^2 - 1} = \frac{c}{t}(t^2 + 1)$$

$$\therefore N \text{ is } (\frac{c}{t}(t^2 + 1), \frac{c}{t}(t^2 + 1))$$

to find T, we need the eqn of tangent at P

$$y - y_1 = m(x - x_1)$$

$$y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$$

$$t^2y - ct = -x + ct$$

$$\therefore x + t^2y - 2ct = 0$$

let $y = x$

$$x + t^2x = 2ct$$

$$x(t^2 + 1) = 2ct$$

$$x = \frac{2ct}{t^2 + 1}$$

$$\therefore T \text{ is } (\frac{2ct}{t^2 + 1}, \frac{2ct}{t^2 + 1})$$

$$(iii) \quad OT = \sqrt{(\frac{2ct}{1+t^2} - 0)^2 + (\frac{2ct}{1+t^2} - 0)^2}$$

$$= \sqrt{2 \cdot \frac{4c^2t^2}{(1+t^2)^2}}$$

$$= \sqrt{2} \cdot \frac{2ct}{1+t^2}$$

$$ON = \sqrt{(\frac{c}{t}(t^2 + 1) - 0)^2 + (\frac{c}{t}(t^2 + 1) - 0)^2}$$

$$= \sqrt{2 \cdot \frac{c^2}{t^2}(t^2 + 1)^2}$$

$$= \sqrt{2} \cdot \frac{c}{t}(t^2 + 1)$$

$$\therefore OT \cdot ON = \sqrt{2} \cdot \frac{2ct}{1+t^2} \times \frac{c}{t}(t^2 + 1) \cdot \sqrt{2}$$

$$= 4c^2$$

$$b) i) b^2 = a^2(1 - e^2)$$

$$16 = 25(1 - e^2)$$

$$\therefore e^2 = \frac{9}{25}$$

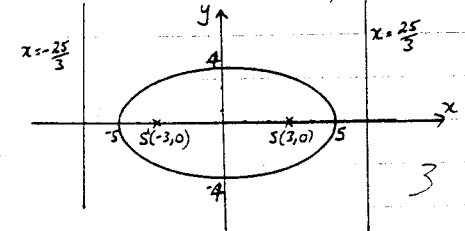
$$e = \frac{3}{5}$$

$$\therefore ae = 5 \times \frac{3}{5} = 3$$

$$\frac{a}{c} = \frac{5}{3/5} = \frac{25}{3}$$

$\therefore$ eccentricity is $\frac{3}{5}$, foci $S(3, 0)$ and $S'(-3, 0)$

and directrices are $x = \frac{25}{3}$, $x = -\frac{25}{3}$



$$(ii) \quad PS + PS' = 2a$$

$$= 10$$

$$\therefore 2 + PS' = 10$$

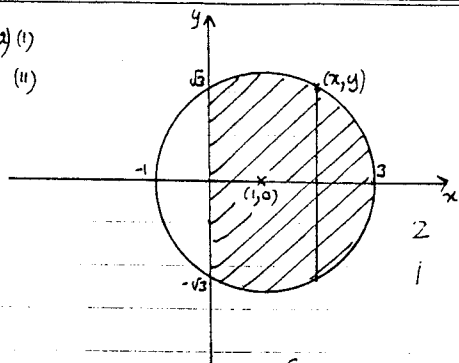
$$PS' = 8$$

$\therefore$ length of PS' is 8 units

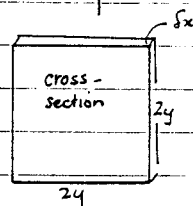
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5.4) (i)

(ii)



(iii)



volume of slice:

$$A = 4y^2$$

$$\therefore \delta V = 4y^2 \delta x$$

$$= 4(4 - (x-1)^2) dx$$

$$= 4(4 - x^2 + 2x - 1) dx$$

$$= 4(3 + 2x - x^2) dx$$

summing all slices

$$\therefore V = \lim_{\delta x \rightarrow 0} \left(4 \sum_{x=0}^3 (3 + 2x - x^2) \delta x \right)$$

$$= 4 \int_0^3 (3 + 2x - x^2) dx$$

$$= 4 \left[3x + x^2 - \frac{x^3}{3} \right]_0^3$$

$$= 4(9 + 9 - 9 - 0 - 0 + 0)$$

$$= 36$$

$\therefore$ volume of solid is 36 units³

$$b) (i) I_n = \int_0^1 x^n e^{2x} dx$$

$$\begin{aligned} \int u v' dx &= uv - \int v u' dx \\ &= \left[x^n \frac{e^{2x}}{2} \right]_0^1 - \int_0^1 \frac{e^{2x}}{2} \cdot n x^{n-1} dx \end{aligned}$$

$$= \frac{e^2}{2} - \frac{n}{2} \int_0^1 x^{n-1} e^{2x} dx$$

$$= \frac{e^2}{2} - \frac{n}{2} \times I_{n-1}$$

$$= \frac{1}{2} (e^2 - n \times I_{n-1})$$

4

$$(ii) I_0 = \int_0^1 x^0 e^{2x} dx$$

$$= \int_0^1 e^{2x} dx$$

$$= \left[\frac{e^{2x}}{2} \right]_0^1$$

$$= \frac{1}{2} (e^2 - 1)$$

$$I_1 = \frac{1}{2} (e^2 - 1 \times I_0)$$

$$= \frac{1}{2} (e^2 - \frac{1}{2} (e^2 - 1))$$

$$= \frac{1}{4} (2e^2 - e^2 + 1)$$

$$= \frac{1}{4} (e^2 + 1)$$

$$I_2 = \frac{1}{2} (e^2 - 2 \times I_1)$$

$$= \frac{1}{2} (e^2 - 2 \times \frac{1}{4} (e^2 + 1))$$

$$= \frac{1}{4} (2e^2 - e^2 - 1)$$

$$= \frac{1}{4} (e^2 - 1)$$

$$I_3 = \frac{1}{2} (e^2 - 3 \times I_2)$$

$$= \frac{1}{2} (e^2 - 3 \times \frac{1}{4} (e^2 - 1))$$

$$= \frac{1}{8} (4e^2 - 3e^2 + 3)$$

$$= \frac{1}{8} (e^2 + 3)$$

$$= \frac{1}{8} (4e^2 - 3e^2 + 3)$$

$$= \frac{1}{8} (e^2 + 3)$$

3

6 a) (a) In Δ 's ABE and DBC

$$\angle ABE = \angle DBC \text{ (data)}$$

$$\angle BAC = \angle BDC \text{ (}\angle\text{'s at circumf. subt same arc are equal)}$$

$$\therefore \Delta ABE \parallel \Delta DBC \text{ (2 prs corr. } \angle\text{'s equal)}$$

(b) In Δ 's ABD and BCE

$$\angle ABD = \angle ABE + \angle EBD$$

$$\angle ECB = \angle DCB + \angle ECD$$

$$\text{But } \angle ABE = \angle DCB$$

$$\therefore \angle ABD = \angle ECB$$

$$\text{Also } \angle BDA = \angle BCE \text{ (}\angle\text{'s at circumf. subt by same arc equal)}$$

$$\therefore \Delta ABD \parallel \Delta ECB \text{ (2 prs corr. } \angle\text{'s equal)}$$

(ii) In Δ 's ABE and BCD

$$\frac{AB}{DB} = \frac{BE}{BC} = \frac{AE}{DC}$$

$$\therefore AB \times DC = DB \times AE \dots (1)$$

In Δ 's ABD and ECB

$$\frac{AB}{EB} = \frac{BD}{BC} = \frac{AD}{EC}$$

$$\therefore AD \times BC = BD \times EC \dots (2)$$

① + ②

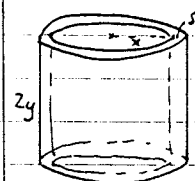
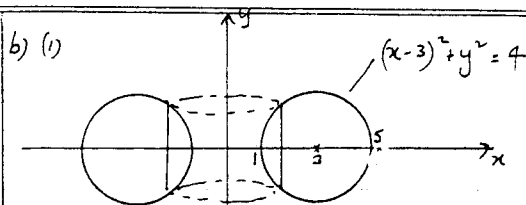
$$AB \times DC + AD \times BC = DB \times AE + BD \times EC$$

$$= DB (AE + EC)$$

$$= DB \times AC$$

$$\therefore AB \times DC + AD \times BC = AC \times BD$$

b) (i)



a typical shell

$$\delta V = \pi ((x + \delta x)^2 - x^2) \cdot 2y$$

$$= 2\pi y (x^2 + 2x\delta x + (\delta x)^2 - x^2)$$

$$= 2\pi y (2x\delta x + (\delta x)^2)$$

We may ignore $(\delta x)^2$ as insignificant

$$\therefore \delta V = 2\pi y \cdot 2x\delta x$$

$$= 4\pi xy \delta x$$

$$\text{since } y^2 = 4 - (x-3)^2$$

$$y = \sqrt{4 - (x-3)^2}$$

$$\therefore \delta V = 4\pi x \sqrt{4 - (x-3)^2} \cdot \delta x$$

summing the shells,

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=1}^5 4\pi x \sqrt{4 - (x-3)^2} \cdot \delta x$$

$$= 4\pi \int_1^5 x \sqrt{4 - (x-3)^2} dx$$

$$(ii) \text{ let } x-3 = 2 \sin \theta \quad \therefore dx = 2 \cos \theta d\theta$$

$$\text{and } x = 3 + 2 \sin \theta$$

$$\therefore V = 4\pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (3 + 2 \sin \theta) \cdot \sqrt{4 - 4 \sin^2 \theta} \cdot 2 \cos \theta d\theta$$

$$= 16\pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (3 + 2 \sin \theta) \cdot \cos \theta \cdot \cos \theta d\theta$$

$$\text{at } x=5, 2 = 2 \sin \theta \quad \therefore \theta = \frac{\pi}{2}$$

$$\text{at } x=1, -2 = 2 \sin \theta \quad \therefore \theta = -\frac{\pi}{2}$$

$$V = 16\pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (3 \cos^2 \theta + 2 \cos^2 \theta \sin \theta) d\theta$$

$$= 16\pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{3}{2} (1 + \cos 2\theta) - 2 \cos^2 \theta \cdot (-\sin \theta) \right) d\theta$$

7

8

6 cont'd.

$$= 16\pi \left[\frac{2}{3} \left(0 + \frac{\sin 2\theta}{2} \right) - \frac{2 \cos^3 \theta}{3} \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$= 16\pi \left(\frac{2}{3} \left(\frac{\pi}{2} + 0 \right) - \frac{2}{3} - \frac{2}{3} \left(-\frac{\pi}{2} - 0 \right) + \frac{2}{3} \right)$$

$$= 16\pi \left(\frac{3\pi}{4} + \frac{3\pi}{4} \right)$$

$$= 24\pi^2 \text{ units}^3$$

Q7(a) $I = \int_1^{\sqrt{x}} \frac{dt}{t}$

$$= [\ln t]_1^{\sqrt{x}}$$

$$= \ln \sqrt{x} - \ln 1$$

$$= \frac{1}{2} \ln x$$

Since $0 < \frac{1}{2} \ln x < \sqrt{x}$

$$\frac{0}{x} < \frac{\frac{1}{2} \ln x}{x} < \frac{\sqrt{x}}{x}$$

$$0 < \frac{\ln x}{x} < \frac{2}{\sqrt{x}}$$

as $x \rightarrow \infty$, $\frac{2}{\sqrt{x}} \rightarrow 0$

$$\therefore 0 < \frac{\ln x}{x} < 0 \text{ as } x \rightarrow \infty$$

i.e. $\lim_{x \rightarrow \infty} \frac{\ln x}{x} = 0$

b) for turning points, $\frac{dy}{dx} = 0$

$$1 - \frac{\ln x}{x} = 0$$

$$\therefore \ln x = 1$$

$$x = e$$

at $x = e$, $y = \frac{\ln e}{e} = \frac{1}{e}$

and $\frac{d^2y}{dx^2} = \frac{2e-3}{e^2} < 0$

$\therefore$ max turning point at $(e, \frac{1}{e})$

for pts of inf. $\frac{d^2y}{dx^2} = 0$ & changes sign

$$\frac{2 \ln x - 3}{x^2} = 0$$

$$\ln x = \frac{3}{2}$$

$$x = e^{\frac{3}{2}}$$

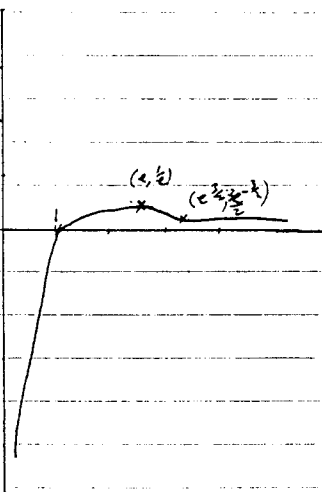
at $x = e^{\frac{3}{2}}$, $y = \frac{\ln e^{\frac{3}{2}}}{e^{\frac{3}{2}}} = \frac{3e^{-\frac{3}{2}}}{2}$

and

x	4	$e^{\frac{3}{2}}$	4.5
$\frac{d^2y}{dx^2}$	$-$	0	$+$

 $\therefore$ changes sign

$\therefore$ pt of inf at $(e^{\frac{3}{2}}, \frac{3e^{-\frac{3}{2}}}{2})$



7 c) (i) target: angle of proj. is 0, speed $-V \text{ ms}^{-1}$

horizontal:

$$\ddot{x} = 0 \quad *$$

$$\dot{x} = \int 0 dt$$

$$= c$$

at $t=0$, $\dot{x} = V \cos \theta$

$$= V$$

$$\therefore \dot{x} = -V \quad *$$

$$x = \int -V dt$$

$$= -Vt + c_1$$

at $t=0$, $x = a$

$$a = 0 + c_1$$

$$\therefore x = -Vt + a$$

$$= a - Vt \quad *$$

(2)

vertical

$$\ddot{y} = -g \quad *$$

$$\dot{y} = \int -g dt$$

$$= -gt + c_2$$

at $t=0$, $\dot{y} = -V \sin \theta$

$$= 0$$

$$\therefore 0 = 0 + c_2$$

$$\therefore \dot{y} = -gt \quad *$$

(3)

$$y = \int -gt dt$$

$$= -\frac{1}{2}gt^2 + c_3$$

at $t=0$, $y = h$

$$\therefore h = 0 + c_3$$

$$\therefore y = -\frac{1}{2}gt^2 + h$$

$$= h - \frac{1}{2}gt^2 \quad *$$

(4)

(ii) shell

$$\ddot{x} = 0$$

$$\ddot{y} = -g$$

$$\dot{x} = U \cos \theta \quad (5)$$

$$\dot{y} = -gt + U \sin \theta \quad (6)$$

$$x = Ut \cos \theta \quad (6)$$

$$y = -\frac{1}{2}gt^2 + Ut \sin \theta \quad (7)$$

at $x = \frac{a}{2}$

$$\frac{a}{2} = Ut \cos \theta \quad \text{From (5)}$$

$$\therefore t = \frac{a}{2U \cos \theta}$$

So at $t = \frac{a}{2U \cos \theta}$

height of target = height of shell

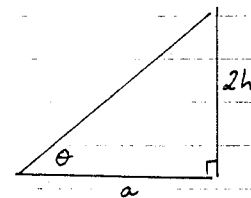
$$-\frac{1}{2}gt^2 + h = -\frac{1}{2}gt^2 + Ut \sin \theta \quad (8)$$

$$h = Ut \sin \theta$$

$$= U \cdot \frac{a}{2U \cos \theta} \cdot \sin \theta$$

$$\therefore \frac{2h}{a} = \tan \theta \quad (9)$$

So the gun is aimed thus:



$\therefore$ gun is aimed at a point $2h$ metres above ground

i.e. aimed h metres above aeroplane

7b(iii) To prove $u = \frac{V}{a} \sqrt{a^2 + 4h^2}$

from (9) $h = \frac{a \tan \theta}{2}$

$\therefore R.H.S = \frac{V}{a} \sqrt{a^2 + a^2 \tan^2 \theta}$

$= \frac{V}{a} \cdot a \sqrt{1 + \tan^2 \theta}$

$= V \sqrt{\sec^2 \theta}$

$= V \sec \theta$

at $x = \frac{a}{2}$

$\frac{a}{2} = a - vt$ from (2)

$\therefore t = \frac{a}{2V}$

also $\frac{a}{2} = \frac{u}{2} \cos \theta$

$\therefore t = \frac{a}{2u \cos \theta}$

$\therefore \frac{a}{2V} = \frac{a}{2u \cos \theta}$

$\therefore \frac{2V}{a} = \frac{2u \cos \theta}{a}$

$\frac{V}{u} = \cos \theta$

$\therefore \sec \theta = \frac{u}{V}$

So $R.H.S = V \cdot \frac{u}{V}$

$= u$

$= L.H.S$

$\therefore u = \frac{V}{a} \sqrt{a^2 + 4h^2}$

8. a(i) $\tan(A+B) = \frac{\sin(A+B)}{\cos(A+B)}$

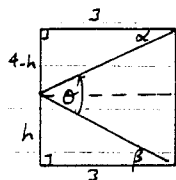
$= \frac{\sin A \cos B + \sin B \cos A}{\cos A \cos B - \sin A \sin B}$

$= \frac{\sin A \cos B}{\cos A \cos B} + \frac{\sin B \cos A}{\cos A \cos B}$

$= \frac{\sin A \cos B}{\cos A \cos B} - \frac{\sin A \sin B}{\cos A \cos B}$

$= \frac{\tan A + \tan B}{1 - \tan A \tan B}$

(ii)



$\theta = \alpha + \beta$

$\tan \alpha = \frac{4-h}{3}$, $\tan \beta = \frac{h}{3}$

$\therefore \tan \theta = \tan(\alpha + \beta)$

$= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$

$= \frac{\frac{4-h}{3} + \frac{h}{3}}{1 - \frac{4-h}{3} \cdot \frac{h}{3}}$

$= \frac{12}{9 - (4-h)h}$

$= \frac{12}{9 - 4h + h^2}$

$\therefore \theta = \tan^{-1} \frac{12}{9 - 4h + h^2}$

$\frac{d\theta}{dh} = \frac{1}{1 + \left(\frac{12}{9 - 4h + h^2}\right)^2} \times -12(9 - 4h + h^2)^{-2} \times (-4 + 2h)$

$= \frac{1}{\frac{(9 - 4h + h^2)^2 + 144}{(9 - 4h + h^2)^2}} \times \frac{-24(h-2)}{(9 - 4h + h^2)^2}$

Q8 cont'd

a) $\frac{d\theta}{dh} = \frac{-24(h-2)}{(9 - 4h + h^2)^2 + 144}$

For max/min values of θ , $\frac{d\theta}{dh} = 0$

ie $-24(h-2) = 0$, $(9 - 4h + h^2)^2 + 144 > 0$

$h = 2$

Check

h	2	2	2
$\frac{d\theta}{dh}$	+	0	-

$\therefore$ max value of θ occurs if $h = 2$

b) period = $2 \times 6\frac{1}{4}$ hrs

$= 750$ mins

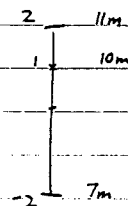
$P = \frac{2\pi}{n}$

$\therefore \frac{2\pi}{n} = 750$

$n = \frac{2\pi}{750} = \frac{\pi}{375}$

difference between high & low tides is 4 m

$\therefore \text{amp} = 2$



since it is SHM,

$\ddot{x} = -n^2 x$

$\therefore x = a \cos(nt + \alpha)$

$= 2 \cos\left(\frac{\pi t}{375} + \alpha\right)$

at $t = 0$, $x = 2$ (at previous H Tide, 2.50 am)

$2 = 2 \cos \alpha$

$\cos \alpha = 1$

$\therefore \alpha = 0$

$\therefore x = 2 \cos \frac{\pi t}{375}$

at $x = 1$, depth of water is 10 m

$\therefore 1 = 2 \cos \frac{\pi t}{375}$

$\cos \frac{\pi t}{375} = \frac{1}{2}$

$t = \frac{375}{\pi} \cdot \cos^{-1}\left(\frac{1}{2}\right)$

$= \frac{375}{\pi} \times \frac{\pi}{3}$

$= 125$ mins, 625 mins, etc

$\therefore$ depth of 10 m next occurs 625 mins

after 2.50 am, ie 1.15 PM

ie pm is the earliest time

(125 mins is before next Low tide $\therefore$ too early)